



AUDITORS • CONTRIBUTORS • PARITY HUNTERS

REFERENCE MANUAL

Compatibility, status, gaps, numerical validation, pine-worker

Open-source Pine Script evaluation framework



CONTENTS

01	Ecosystem
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ECOSYSTEM

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title: "Ecosystem" description: "PYNE, PyneTS, pyne-worker, pyne-agent-worker, AXIS, and HOOX — what each one is and is not."

ECOSYSTEM

Abstract

The HOOX family splits **language**, **chart**, and **execution**. Mixing the names is the most common docs failure: `pyne-worker` is not `pynets`, and AXIS does not place HOOX orders.



Product	Role	Docs
PYNE (<code>hoox-pyne</code> , <code>import pynescrypt</code>)	Python language SoT: parse, LSP, Pro API, <code>Runtime.run</code>	this site
PyneTS (<code>@hoox-sh/pynets</code>)	TypeScript / Bun library + CLI	PyneTS
pyne-worker	Python Cloudflare isolate — <code>POST /run</code>	pyne-worker · HOOX isolate
pyne-agent-worker	NL → Pine (Workers AI)	agent · AXIS plugin
AXIS	Charting PWA (engines call Python)	AXIS
HOOX	Edge trade mesh	HOOX

Conceptual model

```
flowchart TB
    subgraph Language
        PYNE[PYNE Python SoT]
        PYNETS[PyneTS TS library]
        PYNE -->|oracle| PYNETS
    end
    subgraph Evaluate_hosts[Evaluate hosts]
        PRO[Flask Pro API :5002]
        PYW[pyne-worker POST /run]
        PYODIDE[AXIS Pyodide wheel]
        PYNE --> PRO
        PYNE --> PYW
        PYNE --> PYODIDE
    end
    subgraph UI
        AXIS[AXIS PWA]
        AGENT[pyne-agent-worker]
    end
    subgraph Execution
        HOOX[HOOX trade-worker]
    end
    AXIS -->|engine server / proxy| PRO
    AXIS -->|engine pyodide| PYODIDE
    AGENT -->|optional validate| PYW
    PYW -->|StrategyEvent| HOOX
    AXIS -. ->|does not execute| HOOX
```

Invariant: evaluation never requires a proprietary chart host. AXIS is an optional chart. HOOX is an optional execution mesh. PyneTS is an optional TypeScript import.

Interface surface

Published versions (static; a daily cron rewrites the table when a registry moves):



Checkouts (this repo vs sisters)

Checkout	What you open
hoox-sh/pyne	PYNE SoT (hoox-pyne 0.6.7)
hoox-sh/pynets	Standalone @hoox-sh/pynets 0.2.0 (JS compile + stream)
hoox-sh/pyne-worker	Python edge /run
hoox-sh/pyne-agent-worker	NL authoring (not in this tree)
hoox-sh/axis	AXIS PWA
hoox-sh/hoox	Mesh monorepo

Evaluate contract

Flask `POST /run`, pyne-worker `POST /run`, and in-process `Runtime.run` share the **evaluate contract**: `script` + `OHLCV` + `mode` → `plots` / `events` / `alerts`. PyneTS speaks the **library** form of that envelope (`RuntimeResult`), not HTTP.

Internals

Docs are authored in product trees and synced to hoox.sh:

Tree	Public base
pynescrypt/docs/pyne/**	/pyne/docs
axis/docs/**	/axis/docs
hoox/docs/**	/docs

Do not author in `hoox-landing-page/content/*` — that is a sync artifact.

Invariants & edge cases

1. **Python wins** language semantics. PyneTS does not invent TradingView behaviour.
2. **AXIS ≠ engine**. AXIS never embeds a closed interpreter; engines are plugins.
3. **AXIS ≠ execution**. Drawing a strategy on AXIS does not call `trade-worker`.
4. **Alerts ≠ orders**. `alert()` webhooks are not `StrategyEvent` forwards unless you point them at the HOOX gateway yourself.
5. **Do not** `pip install pyne` or `pip install pynescrypt` for this project. Dist name is **hoox-pyne**.
6. **Do not** `npm install pynets` for this project. The published package is **@hoox-sh/pynets** from the standalone **hoox-sh/pynets** repo. Python Runtime remains the oracle.
7. **pine-worker** is a leftover TypeScript Cloudflare experiment (**hoox-sh/pine-worker**). It is not a product evaluate host and has no pages on this site.



Worked examples

Pick a surface

You want	Use
Parse / lint / LSP on a laptop	<code>pip install "hoox-pyne[lsp]"</code>
Embed evaluate in TypeScript	<code>bun add @hoox-sh/pynets</code>
HTTP evaluate at the edge	Deploy pyne-worker
Upload then keep running (Flask/VPS/container)	<code>PYNE_RUNNER=1</code> — hosted runner
Chart + editor	AXIS PWA + Flask or Pyodide
Live CEX orders from strategy events	<code>pyne-worker</code> <code>TRADE_SERVICE</code> → HOOX
Chat → script	pyne-agent-worker

Failure modes

Mix-up	What goes wrong
Opening <code>pyne-worker/</code> or <code>pine-worker/</code> inside PYNE	Directories do not exist — clone the sister repos; TS library work is <code>@hoox-sh/pynets</code>
Expecting AXIS to flatten a position	No trade-worker hop
Pointing AXIS at pynets as an engine	No such engine. Use Flask / Pyodide / Worker proxy
<code>pyne run</code> vs <code>Runtime.run</code> vs <code>pynets run</code>	Different defaults — see modes

See also

- **PYNE index**
- **PyneTS**
- **pyne-worker**
- **pyne-agent-worker**
- **Modes**
- **Evaluate contract**
- **AXIS**
- **HOOX**
- **HOOX live trading**
- **AXIS and HOOX**

COMPATIBILITY

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title: "Compatibility guarantee" description: "Grouped map of what PYNE implements for Pine Script™ v5/v6 — working, partial, residual, and by-design out of scope. hoox-pyne 0.6.7."

COMPATIBILITY GUARANTEE

Abstract

PYNE implements Pine Script™ **v5/v6 language core** as an inspectable pipeline: parse → AST → bar-loop evaluate. The map below is the product-facing compatibility surface for **hoox-pyne 0.6.7**. It is **not** TradingView® platform identity (chart host, proprietary data, editor-only UI) and **not** bit-identical results vs the hosted platform on every script or bar.



Name-level dispatch lives in [Pine v6 surface](#) (docs/pine_v6_full_surface_inventory.md). Gaps: [Missing features](#). Checkmarks: [Implementation status](#).

Conceptual model

```
flowchart TB
    SRC[".pyne / .pine"] --> PARSE["Parser + ASDL AST"]
    PARSE --> LOOP["Bar-loop Runtime"]
    LOOP --> INTERP["interpret – full surface"]
    LOOP --> COMP["compile – Numba / object"]
    INTERP --> OUT["plots · events · alerts · drawings"]
    COMP --> OUT
    OUT --> HOSTS["CLI · Pro API · workers · AXIS"]

    subgraph working [Working]
        PARSE
        INTERP
        TA["ta.* incremental hot path"]
        STR["strategy broker"]
        COL["array / matrix / map"]
        DRAW["plot / line / box / label"]
    end

    subgraph partial [Partial – honest NA or host-bound]
        REQ["request.* mock / same-symbol"]
        COMP
        PAR["interp ↔ compile plot tail"]
    end

    subgraph oos [By design – not this runtime]
        TV["TV chart / editor UI"]
        FEED["foreign live fundamentals"]
        PIX["pixel renderer"]
    end

    LOOP --> TA
    LOOP --> STR
    LOOP --> COL
    LOOP --> DRAW
    LOOP --> REQ
```

Legend

Mark	Meaning
Working	Implemented, usable, covered by first-party tests
Partial	Callable but mock, host-bound, or incomplete vs hosted Pine
Residual	Known open hole on an otherwise landed surface
Out of scope	Intentionally not a TV platform clone

Dispatch snapshot (2026-07-25): **640** callables, **0 missing** vs the public TV v6 function list (434 symbols). A registered name is not the same as hosted-platform semantics.



Language, series, TA hot path, collections, strategy broker, drawings, alerts, libraries, interpret

`request.\*` feeds (honest `na`), compile eligibility, compile alerts, PyneTS vs Python oracle.

Plot MISMATCH corpus tail (P1p), `PYNE\_SERIES\_RING` default off.

Pixel chart, editor UX, foreign live data, bit-identical TV bars.

Interface surface

Corpus (set01–04 · 2477 scripts · 2026-08-09)

Open-source sets, **not shipped** in git. Not a TradingView® platform score.

Suite	Rate	Notes
Parse + unparse	99.96% (2476/2477)	Residual: one intentional invalid line-wrap demo
Runtime interpret (50 bars)	100% excl. EXPECTED_FAIL	2466 OK + 11 listed demos
set01 Runtime	249/249	Interpret (and prior compile sweep)

EXPECTED_FAIL are path-listed demos (`runtime.error` library guards, lower-TF security, pathological loops) — not silent suppression.



Grouped map

```
| Area | Status | Notes |
| --- | --- | --- |
| ANTLR4 grammar v5/v6 | Working | Multiline strings, soft keywords, bitwise, typed UDF returns ('F
| Parse → unparse | Working | Structural fidelity; whitespace may differ |
| `var` / `varip` / `:=` | Working | Execution-site init; start-of-bar carry |
| Functions, methods, type annotations | Working | |
| Control flow, `switch`, `once`, enums | Working | Strict bool; `na` is false in conditions. `once
| UDTs + `.new` | Working | Field defaults on compile (0.3.8) |
| Libraries `export` / `import ns/Name/ver` | Working | In-process `register_library_source`; `Runt
| String interpolation, tuple unpack, method chaining | Working | |
| Series `[]` history | Working | OOB / negative → `na` (not wrap) |
| `timeframe.change` | Working | Intraday fixed-width buckets; D/W/M on the exchange calendar (0.6.
```

```
| Area | Status | Notes |
| --- | --- | --- |
| `pynescrypt.runtime.Runtime` | Working | Package SoT; `backend.runtime` re-exports |
| Modes `interpret` / `compile` / `auto` | Working | Defaults differ by surface – [modes](/pyne/doc
| Series caps | Working | `PYNE_SERIES_CAP` default **on** |
| Incremental `ta.*` | Working | Default **on**;; `PYNE_TA_INCREMENTAL=0` full recompute. **0.3.12**
| Derived OHLCV skip | Working | Unused `hl2`/`hlc3`/`ohlcv4`/`tr` not written; `ta.vwap` still upda
| `timeout_seconds` | Working | Library / edge / Flask `/run` + `/run/batch` (omit = no budget) |
| Ring buffer | Residual | `PYNE_SERIES_RING` default **off** |
```

```
| Area | Status | Notes |
| --- | --- | --- |
| MAs | Working | `sma`/`ema`/`rma`/`wma`/`hma`/`vwma`/`kama`/`dema`/`tema`/`alma` incremental |
| Oscillators | Working | `rsi`/`macd`/`stoch`/`cci`/`cmo`/`tsi`/`roc`/`wpr`; **0.3.12** incrementa
| Volatility / structure | Working | `atr` (Wilder `rma(tr)`), `bb`, `kc`, `stdev`, `highest`/`lowe
| Volume | Working | Incremental `obv`/`wad`/`wvad`/`cmf`/`klinger`/`mfi`/`vwap`/`nvi`/`pvi` |
| Supertrend | Working | Locked mid±factor·ATR (inc ≡ compile ≡ Numba). TV band ratchet is out of s
```

```
| Area | Status | Notes |
| --- | --- | --- |
| `array.*` | Working | Negative indices; UDT `sort_field` + `binary_search*` |
| `matrix.*` | Working | LA (`det`/`inv`/`eigen`...), predicates, `sort_field` |
| `map.*` | Working | |
```

```
| Area | Status | Notes |
| --- | --- | --- |
| `entry` / `close` / `cancel` / events | Working | `StrategyEvent` stream |
| `exit` stop/limit + OCA | Working | OHLC path, not tick path |
| `profit` / `loss` | Working | Ticks × mintick from entry avg |
| `from_entry` / `qty_percent` | Working | Interpret + compile |
| Trail | Working | `trail_offset` / `trail_points` / `trail_price` on OHLC high/low (dual-host). 1
| Risk cascade | Working | `allow_entry_in`, max size/drawdown/cons-loss-days/intraday |
| Open/closed trade fields + MAE/MFE | Working | Extremes from bar high/low |
| Pending-fill average (pyramiding ≤ 0) | Working | F2 |
| Licensed broker / exchange fills | Out of scope | In-process model only |
```



```
| Area | Status | Notes |
| --- | --- | --- |
| `plot` / `plotshape` / `plotchar` / `plotarrow` / `hline` / `bgcolor` / `barcolor` / `fill` / `pl
| `line` / `box` / `label` / `table` / `polyline` / `linefill` | Working | Coordinates snapshotted
| `max_*_count` GC | Working | Interpret registry |
| `force_overlay` | Working | Export payload |
| Pixel paint | Out of scope | AXIS / clients consume series + drawings |
| Compile `hline`/`fill`/`bgcolor`/`plotshape` keys | Working | First-party key sets match interpre
```

```
| Area | Status | Notes |
| --- | --- | --- |
| Same-symbol simple OHLCV / HTF | Working | Allowlisted `ta.sma`/`ema`/`rsi`/`atr`/`wma`/`rma` on
| Foreign / complex `request.security` | Partial | Resolves to **`na`** (no invented chart-close) |
| `request.footprint` / financial / economic | Partial | Mock / feed-scaled; not a live vendor |
| `input.*` including `active` / `enum` | Working | Values + metadata |
| Bid/ask omitted | Partial | `na` when host does not supply |
| Real multi-symbol market data (B1) | Out of scope | Needs a host adapter |
```

```
| Area | Status | Notes |
| --- | --- | --- |
| `alert()` / `alertcondition()` | Working | Frequency rules; `/run` export |
| L2 webhooks | Working | `webhook_url` / `ALERT_WEBHOOK_URL` |
| Compile alerts | Partial | Interpret is the alert oracle |
| Numba + object-mode compile | Working | `mode=auto` falls back; `import` / `request.*` / `inputs=
| Warm compile / disk IR | Working | Prewarm CLI/API; cache meta v16 |
| Interp ↔ compile plot parity | Working | Harness + first-party goldens + official TV builtins **6
| CLI / LSP / Pro API / Docker | Working | `pyne` / `pyne-lsp`; Flask `/run` |
| Flask `/run` `libraries` | Working | Max 32; also `/run/batch` |
| Flask `timeout_seconds` | Working | Optional on `/run` and `/run/batch`; omit / `≤ 0` = no timeou
| `timeframe.change` | Working | Intraday fixed-width; D/W/M exchange calendar. Interpret + compile
| `POST /optimize` / `pyne optimize` | Working | Strategy `input.*` search – not a Pine builtin |
```

```
| Surface | Status | Notes |
| --- | --- | --- |
| Python Runtime | Working | Language oracle |
| PyneTS `@hoox-sh/pynets` 0.2.0 | Partial | Interpret + JS compile on npm; Python Runtime is the c
| `pynets/` submodule in this repo | Partial | Pin **v0.2.0** – same surface as npm (interpret + JS
| pyne-worker | Working | Thin Python CF host; vendors engine (may lag 0.3.16). Docs: [pyne-worker]
| pyne-agent-worker | Working | NL authoring; optional `/run` validate. Docs: [agent](/pyne/docs/ag
```



Residual and out of scope (compact)

Item	Class
Interpret ↔ compile value <code>MISMATCH</code> (optional set0x community corpus)	Residual (P1p official builtins closed)
<code>PYNE_SERIES_RING</code> default-on	Residual (flagged off)
Pixel chart / editor word-wrap / TV UI	Out of scope
Foreign live fundamentals without a feed	Out of scope
TV Supertrend band ratchet	Out of scope
Tick-path trail / broker	Out of scope
Bit-identical every recursive smoother vs TV	Out of scope
Parallel bars / whole-script vectorization	Out of scope

```
flowchart LR
    subgraph W[Working]
        L[Language]
        H[Interpret host]
        T[ta.* hot path]
        C[Collections]
        S[Strategy]
        D[Draw / plot]
        A[Alerts]
    end
    subgraph P[Partial]
        R[request feeds]
        Y[PyneTS vs Python]
    end
    subgraph X[Residual]
        M[P1p MISMATCH tail]
        RING[series ring off]
    end
    subgraph O[Out of scope]
        U[TV UI / pixels]
        V[foreign live data]
    end
```

Dual-host plot parity

Same script + OHLCV under `mode="interpret"` and `mode="compile"` ; series compared with nan-aware allclose (`rtol=1e-5`, `atol=1e-6`).

Artifact	Role
<code>scripts/compare_interp_compile.py</code>	Corpus harness → <code>.cache/interp_compile_parity.json</code>
<code>tests/test_interp_compile_parity.py</code>	Always-on smoke
<code>tests/test_first_party_ta_goldens.py</code>	ATR / Supertrend / Keltner dual-host

Known **contract** differences (not silent value bugs):

1. Foreign / complex `request.*` → `na` on both hosts when no feed.



2. First-party `hline` / `fill` / `bgcolor` / `plotshape` keys match; harness `--ignore-hline-keys` / `--ignore-fill-keys` remain optional for leftover corpus noise.
3. Pivot-starved Auto Fib → matching `runtime.error` (`both_error_same`), not invented pivots.

Internals

Artifact	Role
<code>docs/compatibility_guarantee.md</code>	Historical long-form; this page is the map
<code>COMPATIBILITY.md</code>	Repo-root distillation of this map
<code>docs/pine_v6_full_surface_inventory.md</code>	Name-level dispatch tables
<code>docs/known_divergences.md</code>	Semantic deltas vs public Pine docs
<code>tests/fixtures/parity/</code>	First-party strategy fixtures

Invariants & edge cases

1. **Parse success ≠ evaluate success.** Round-trip can still depend on `request.*` data.
2. **Dispatch success ≠ hosted semantics.** 0 missing vs the public function list does not mean TV-identical bars.
3. **Interpret is the oracle.** Compile matches where eligible; `mode=auto` falls back.
4. **Plots are not a renderer.** Registry + series export; pixels are AXIS / clients.
5. **Floating-point is statistical** for some recursive smoothers — [Numerical validation](#).
6. **Libraries** need in-process registration; there is no TV CDN.

Worked examples

Round-trip

```
from pynescrypt.ast.helper import parse, unparse

src = open("strategy.pine").read()
assert unparse(parse(src)) # structural; whitespace may differ
```

Interpret ↔ compile

```
python scripts/compare_interp_compile.py --bars 1000 --limit 50
python scripts/compare_interp_compile.py --ignore-hline-keys --ignore-fill-keys
pytest tests/test_interp_compile_parity.py -q
```



Failure modes

Observation	Read as
Parse error on a new TV release-notes feature	Surface gap — Missing features
Numerical drift on a custom smoother	Bar-mode vs list-mode; <code>na</code> ; known divergences
Interpret vs compile series mismatch	Foreign <code>request.*</code> , UDF last-assign <code>na</code> , leftover corpus keys
Strategy equity $\neq$ TV	Commission / slippage / OHLC vs tick path
<code>import</code> unresolved	Library not in <code>libraries=</code> / registry
<code>/run 400 UNKNOWN_FIELDS</code>	Extra key not in <code>RUN_SCHEMA</code>

See also

- [Implementation status](#)
- [Missing features](#)
- [Pine v6 surface inventory](#)
- [Numerical validation](#)
- [Runtime modes](#)
- [Roadmap](#)

IMPLEMENTATION STATUS

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title: "Implementation status" description: "Checklist-style status of Pine Script™ series, strategy, builtins, and language constructs in PYNE."

IMPLEMENTATION STATUS

Abstract

This annex summarizes the **feature matrix** maintained in `docs/pinescript_implementation_status.md`: a large    inventory of series variables, strategy fields, and related surfaces. It is the human-readable twin of the **full surface inventory** (dispatch-oriented, auto-countable).

Status here is **implementation judgment**, not marketing completion percentage.



Conceptual model

```
flowchart LR
    TV[TradingView v6 surface] --> INV[Inventory docs]
    INV --> DISP[Evaluator dispatch map]
    INV --> CTX[Default context series]
    INV --> AST[Grammar / ASDL]
    DISP --> TESTS[pytest]
    CTX --> TESTS
    AST --> TESTS
```

Interface surface

Legend

Mark	Meaning
	Implemented and usable
	Partial / stub / mock
	Missing
	By design out of scope
	Editor/platform N/A



Component rollup

Component	Posture (2026-07 consolidation)
Parser (ANTLR4)	Complete for v5/v6 core + multiline / export / once (0.4.4)
Evaluator	Full bar-loop with var/varip, reassignment
Builtins	224+ historical count; 640 callables on live dispatch inventory (2026-07-25)
TA	150+ ta.* keys in inventory
Collections	array / matrix / map complete on exercised surface
Strategy events	StrategyEvent , parity corpus, risk enforcement
Drawing / plot	Registry effects; AXIS is external
Linter	Present (pynescrypt lint)
Data providers	Mock, Yahoo, AlphaVantage, CCXT + datafeed wiring
LSP	Diagnostics, completion, hover, format, symbols, defs/refs, ...
PyneTS	@hoox-sh/pynets 0.2.0 (npm + this repo's pynets/ pin v0.2.0) — interpret + JS compile. Python Runtime remains the oracle — see PyneTS
pyne-worker	Python Cloudflare evaluate host (sister repo) — docs
pyne-agent-worker	NL authoring host (sister repo) — docs
Numba compile path	MVP + object-mode fallback; 0.3.16–0.3.17 UDT/switch/matrix/drawing/round/color residual recovery
Incremental ta.*	Default on (PYNE_TA_INCREMENTAL); 0.3.10 volume obv / wad / cmf / klinger ; 0.3.11 nvi / pvi ; 0.3.12 aroon / dpo / donchian / kst
Interpret bar-loop	0.3.10 dispatch inlining + unused derived-series skip

Series & context (illustrative families)

Fully listed in the source doc; representative groups:

- **Price:** **open** / **high** / **low** / **close** / **volume** / **hl2** / **ohlc4** / ...
- **Time:** **year ... second** , **time_close** , **timenow** , ...
- **Barstate / chart:** **bar_index** , **barstate.*** , **last_bar_***
- **syminfo.*** ticker, mintick, session, fundamentals fields tracked as 
- **session.*** / **dividends.*** / **earnings.*** context fields tracked as 
- **strategy.*** position, trades, equity, risk, OCA, commission constants

July 2026 enhancements called out in source

- Full strategy event emission with bar context
- **strategy.long** / **strategy.short** constants
- **var** / **varip** + **:=** reassignment
- Parity fixtures for cross-implementation validation
- Risk helpers and performance series expansion



Internals

Path	Role
docs/pinescript_implementation_status.md	Exhaustive checklist (do not paste wholesale here)
docs/pine_v6_full_surface_inventory.md	Dispatch-centric inventory
src/pynescrypt/ast/evaluator/builtins/	Per-namespace implementations
scripts/regenerate_inventory_summary.py	Inventory regeneration aid

Invariants & edge cases

1. **Checklist lag**: a  may trail a commit by a day; prefer tests when shipping.
2. **Context fields may be defaults** rather than live exchange feeds.
3. **Partial stubs still appear callable** — inventory marks  when docstrings/heuristics say mock.
4. **Namespace counts ≠ quality**. Many drawing setters are thin mutators; TA is heavier.

Worked examples

Smoke a strategy series field

```
from pynescrypt.ast.helper import parse
# evaluate via Runtime / library API with OHLCV — see runtime docs
```

Regenerate inventory summary

```
python scripts/regenerate_inventory_summary.py
```

Failure modes

Symptom	Action
Doc says  , runtime AttributeError	File bug; fix dispatcher or demote status
Partial mock accepted silently	Assert data_feed wiring in tests
Status doc conflicts inventory	Prefer live dispatch + tests

See also

- [Compatibility map](#) — grouped working / partial / residual / out of scope
- [Missing features](#)
- [Pine v6 surface](#)
- [Roadmap](#)

MISSING FEATURES

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title: "Missing features" description: "Known gaps and closed Pine Script™ v6 items — 0.6.7 interpret dispatch and compile ta.uo; 0.6.6 last-sample TA and convert/for-n safety; 0.6.5 convert-to-v6 UDF-safe prefixes; 0.6.4 convert-to-v6; 0.6.3 hosted runner; 0.6.2 P1p official builtins 0 MISMATCH."

MISSING FEATURES

Abstract

“Missing” in PYNE is a **moving frontier**. As of **2026-09-23** (`hoox-pyne` **0.6.7**), core v6 support remains very high (~99%+ for core language), with post-launch TV items closed (multiline strings, library `export const`, footprint mocks, risk enforcement, matrix LA, compile-mode strategy, drawing GC, ...) **plus** open-source corpus



parse/Runtime hardening, **incremental bar-mode TA** (including 0.3.10–0.3.12 kernels), **alert engine + L2 webhooks**, **series caps**, **warm-compile**, **pending-fill** averaging, **timeframe.change** (intraday buckets + 0.6.0 D/W/M exchange calendar), **package Runtime SoT** (`pynescrypt.runtime` ; `backend.runtime` is a shim), **compile object-mode residual recovery** (UDF locals, nopython `None` /unicode, matrix handles, UDT returns, `x = switch`, drawing copy, `int(na)` , `round()` extras, import stubs, color type dispatch, nested UDT methods), **type-aware intraday risk enforcement on both brokers**, intrabar **var** rollback on realtime ticks, **request.security** barmerge on HTF resample paths, and **linter legacy-rule retirements**. Remaining gaps cluster into **optional set0x community-corpus plot parity (P1p official builtins closed)**, **by-design platform differences**, and **editor-only features**. Incremental `ta.nvi` / `ta.pvi` / `aroon` / `dpo` / `donchian` / `kst` and Supertrend $\text{mid} \pm \text{factor} \cdot \text{ATR}$ goldens have landed.

Authoritative narrative: `docs/missing_features.md` . Canonical IDs: `docs/ROADMAP.md` (2026-08-17).

Conceptual model

```
flowchart TB
    subgraph Closed_recently
        ML[Multiline strings]
        EX[export const / types / enums]
        FP[request.footprint mocks]
        MAT[matrix LA surface]
        STR[strategy runtime depth]
        CORP[Corpus sanitize + grammar]
        PERF[Incremental bar-mode TA]
        ALERT[Alerts + L2 webhooks]
        R7[R7 series caps / warm compile / F2]
    end
    subgraph Remaining
        RD[Real multi-symbol request.* feeds]
        ED[Editor-only UX]
        P1p[optional set0x community-corpus plot parity]
    end
    TV[TV release notes] --> Closed_recently
    TV --> Remaining
    CORP --> TAIL
```

Interface surface

Recently completed (do not re-open as “missing”)

Summarized — full prose in source doc:



Area	Notes
Multiline <code>"""</code> / <code>'''</code>	Lexer + LexerBase + unparser
Library <code>export const</code> + types/enums	Parse/AST/runtime registry
UDT <code>sort_field</code> on array/matrix sort	Implemented
UDT <code>sort_field</code> on <code>array.binary_search*</code> (August 2026)	Implemented (interpret + compile, 0.3.6)
once conditional structure (August 2026)	Implemented in 0.4.4 — parse/unparse, interpret (confirmed-bar latch + realtime re-fire), compile historical flags. Statement-only.
<code>input.*.active</code>	Metadata-driven
Dynamic <code>request.*</code>	Shared resolvers; mock/feed scaling
Enums runtime + <code>input.enum</code>	Type kind + LSP partial
Strategy open/closed trades, risk, OCA, commission	Golden tests
Drawing <code>*.all</code> , plot registry, <code>max_*_count</code> GC	Effects recorded; GC on package + Pro + AXIS
Numba compile + object mode + warm-compile (H2)	Disk IR cache, prewarm API/CLI, deploy defaults
Compile object-mode residuals (0.3.16–0.3.17)	UDF locals / free-series, nopython <code>None</code> / <code>timestamp</code> , matrix handles, UDT returns, <code>x = switch</code> , drawing copy, <code>pine_int(na)</code> , sanitize chrome, <code>round()</code> extras, import stubs as <code>na</code> , color type dispatch, nested UDT methods
Official matrix LA & predicates	<code>det/inv/eigen/...</code>
0 missing vs public TV v6 function reference list (434 symbols, 2026-07-25)	
Runtime <code>_pine_defs_locked</code> + append-only <code>current_series</code>	Host hygiene (backend + pyne-worker)
Incremental <code>ta.*</code> (MAs, oscillators, <code>bb</code> / <code>kama</code> / <code>stochrsi</code> , volume <code>obv</code> / <code>wad</code> / <code>cmf</code> / <code>klinger</code> / <code>nvi</code> / <code>pvi</code>)	Bar mode; <code>PYNE_TA_INCREMENTAL=0</code> to disable
Series caps (T1)	<code>PYNE_SERIES_CAP</code> default ON + goldens
Pending-fill averaging (F2)	Pyramiding ≤ 0 ; interpret + compile broker goldens
Alert engine + L2 webhooks (L2)	Dual-host <code>/run export</code> , <code>ALERT_WEBHOOK_URL</code> / <code>webhook_url</code>
request.security foreign-na policy	Foreign/complex → <code>na</code> both backends; same-symbol OHLCV passthrough
Plot parity harness (P1p)	<code>scripts/compare_interp_compile.py</code> , first-party goldens + official TV builtins 0 MISMATCH
<code>fill()</code> export for AXIS	series + <code>plot_meta</code> / compile <code>__drawings</code>



Still incomplete / by design

Item	Status posture
Real (non-mock) multi-symbol <code>request.*</code> market data (B1)	⚙ By design for library core; feeds optional. Foreign-na policy landed (no invent).
<code>request.security</code> foreign data	⚙ Not filled with real series — emits <code>na</code> for foreign/complex (honest). Same-symbol simple OHLCV may lower.
Auto Fib / pivot-heavy scripts	⚙ Need real pivot structure (or registered ZigZag lib). Flat synthetic bars → intentional <code>runtime.error</code> (insufficient pivots); both hosts should agree (<code>both_error_same</code> in parity harness).
Pixel chart host	External (AXIS / clients)
Some unlimited-history / trim edge cases	High-level support; exotic TV behaviors may differ
Editor word-wrap / UI-only release notes	☐ N/A
PyneTS full builtin parity (L3)	⚙ interpret + JS compile landed in <code>@hoox-sh/pynets</code> 0.2.0; Python remains oracle. Open rows: PyneTS missing features
Bit-identical every recursive smoother vs TV	Numerical bounds, not bits
Dual Runtime host copies (H1)	✅ largely done — package <code>pynescrypt.runtime</code> SoT + <code>backend.*</code> shims + pyne-worker thin wrap. Residual = worker-only extras (logs/profile, CF first-plot), not a forked bar loop
Warm-compile product path (H2)	✅ SLOs / prewarm / deploy IR cache defaults (2026-08)
Runtime residual (C1)	✅ set01–04 Runtime interpret 100% excl. EXPECTED_FAIL (2026-08-09); third-party corpus not shipped
Series cap (T1)	✅ <code>PYNE_SERIES_CAP</code> default ON
Residual TA inc (T2)	✅ R7 bb/kama/cmo/stochrsi + wma/hma/linreg; 0.3.10 <code>obv</code> / <code>wad</code> / <code>cmf</code> / <code>klinger</code> ; 0.3.11 <code>nvi</code> / <code>pvi</code> ; 0.3.12 <code>aroon</code> / <code>dpo</code> / <code>donchian</code> / <code>kst</code>
Pending-fill (F2)	✅ R7 interpret + compile goldens
Plot parity residual (P1p)	✅ Harness + first-party goldens + official TV builtins 0 MISMATCH ; optional set0x community corpus unmeasured
Compile leftover library stubs	⚙ Some <code>float(None)</code> / UDF arity on missing <code>import</code> libraries; INV expected-fail fixtures stay fail
TV Wilder RMA-ATR / supertrend (F1)	✅ interpret ATR is <code>ta.ima(ta.tr)</code> ; Supertrend $\text{mid} \pm \text{factor} \cdot \text{ATR}$ locked. TV ratchet is out of scope
Alert webhooks (L2)	✅ Pro API + pyne-worker; see Alerts

Post-v6 launch themes (historical gap list)

v6 launch items (dynamic requests, strict bool, enums, polylines, `log.*`, negative indices, `truediv`, ...) are largely supported or intentionally stubbed where platform-bound.

2025–2026 monthly TV updates introduced footprint types, `active` inputs, multiline strings, `export const`, UDT sort fields — tracked and largely closed in this repo's July 2026 work.



Corpus + performance snapshot (2026-08-09)

Open-source set01–04 (2477 scripts; measured locally, **not shipped** in-repo):

Metric	Value
Parse + unparse	99.96% (2476 / 2477) — residual: 1 intentional invalid line-wrap docs demo
Runtime interpret	100% excl. EXPECTED_FAIL — 2466 OK + 11 intentional demos
set01 Runtime	249 / 249 (100%)
EXPECTED_FAIL class	library <code>runtime.error</code> demos, lower-TF security guards, pathological loops, truncated mid-call scrapes
<code>ta_sma</code> bar loop (≈3.2k bars)	~9.5× vs full recompute
<code>ta_atr</code> bar loop	~10×
macd+atr+rsi+sma combo	~8.5×

Not TradingView® platform parity. OK% excludes only path-listed intentional demos (never soft-suppress unknown failures).

Plans: `.opencode/plans/2026-07-28-runtime-performance.md` . Skill: `.grok/skills/pynescrypt-perf/` . Golden: `tests/test_ta_incremental.py` , `tests/test_series_cap.py` .

Internals

Path	Role
<code>docs/missing_features.md</code>	Living gap analysis
<code>tests/test_v6_features.py</code>	v6 feature coverage
<code>tests/test_pine_surface_gaps.py</code>	Surface gap probes
<code>tests/test_ta_incremental.py</code>	Incremental TA ≡ full recompute
<code>tests/test_series_cap.py</code>	Series cap goldens (T1)
<code>scripts/compare_interp_compile.py</code>	Plot parity harness (P1p)
<code>docs/pine_v6_full_surface_inventory.md</code>	Full name-level inventory
<code>resource/*.g4</code> + builder	Grammar-level feature work
<code>.../technical_submodules/core.py</code>	<code>_sma_inc_update</code> , <code>_macd_inc_update</code> , <code>_atr_inc_update</code> , ...

Invariants & edge cases

1. Closing a syntax gap requires grammar + builder + unparser + tests, not only an evaluator stub.
2. Mock data can green tests while production feeds differ — declare data source explicitly.
3. Compile does not fill foreign `request.security` data — multi-asset plots stay `na` on compile unless same-symbol simple OHLCV; do not treat invent-as-close as a fix.
4. Auto Fib needs pivots — without swing structure (or ZigZag library registration), expect structured insufficient-pivot errors, not fib lines.
5. “0 missing vs reference list” ≠ identical semantics for every edge case.



- 6. **Perf changes must keep bar-by-bar semantics** — no whole-script vectorization; prefer golden tests vs current oracle.
- 7. Update this annex when `missing_features.md` changes major status lines.

Worked examples

Detect multiline string support

```
from pynescrypt.ast.helper import parse, unparse
src = '''//@version=6
indicator("m")
s = ""a
b""
plot(1)
'''
tree = parse(src)
assert '""' in unparse(tree) or '"""' in unparse(tree)
```

Footprint mock call

Prefer tests in `tests/test_v6_features.py` as executable specification for `request.footprint` methods.

Incremental TA parity smoke

```
.venv/bin/python -m pytest tests/test_ta_incremental.py -q
# Disable hot path if needed:
PYNE_TA_INCREMENTAL=0 python -c "from pynescrypt.runtime import Runtime; ..."
```

Failure modes

Symptom	Likely gap class
Lexer error on triple quotes	Stale generated lexer (should be fixed on main)
Import member missing	Library export registration
Empty footprint rows	Mock seed / data_feed not configured
Risk not blocking entry	Old runtime path; ensure risk kwargs applied
Runtime TIMEOUT on method-heavy scripts	Missing <code>_pine_defs_locked</code> on host (fixed 2026-07-28)
Corpus PARSE_FAIL mid-block	Truncated scrape — not a grammar hole

See also

- [Implementation status](#)
- [Pine v6 surface](#)
- [Roadmap](#)
- [Compatibility](#)
- [Technical analysis \(ta.*\)](#)



- Alerts
- Interpret ↔ compile parity

PINE V6 SURFACE

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title: "Pine v6 surface inventory" description: "Summary of the full Pine Script™ v6 surface inventory — dispatch counts, namespaces, status schema — with pointer to the large source document."

PINE V6 SURFACE INVENTORY

Abstract

The **full surface inventory** enumerates every registered evaluator builtin, major series/variable, language construct, and known gap against Pine Script™ v6. It is deliberately large (tens of thousands of tokens when fully expanded). **This page is the navigable summary**; the exhaustive tables live in-repo at:

`docs/pine_v6_full_surface_inventory.md`



Generated snapshot referenced here: **2026-07-25**, from live `NodeLiteralEvaluator` dispatch, `builtin_metadata.json` , and design docs.

Do **not** paste the entire inventory into product docs — regenerate and link.

Conceptual model

```
flowchart TB
    subgraph Parse
        SRC[Pine source] --> LEX[Lexer]
        LEX --> PAR[Parser]
        PAR --> BLD[Builder]
        BLD --> AST[ASDL AST]
    end
    subgraph Eval
        AST --> VIS[NodeLiteralEvaluator]
        VIS --> BI[Builtin dispatch map]
        BI --> NS[ta / strategy / array / ...]
    end
    BI --> INV[Inventory rows]
```

Interface surface

Status schema

Status	Definition
 implemented	Handler/parser path exists and is usable
 partial	Mock/stub, incomplete semantics, or metadata-only
 missing	Expected on TV v6; not resolving
 by design	Intentionally not full TV platform
 N/A	Editor/UI-only

Each inventory row also carries: **Name**, **Namespace**, **Kind**, **Metadata** (LSP JSON present?), **Source** (e.g. `dispatch`), **Notes**.

Summary counts (2026-07-25)

Metric	Count
Dispatch builtins (callable)	640
Dispatch partial-heuristic (stub/mock docstring)	8
Top-level namespaces	60
Official TV v6 function reference list	434 symbols — 0 missing in dispatch



Top namespaces by dispatch keys

Namespace	Count
ta	159
matrix	74
strategy	70
array	56
box	31
label	28
table	26
math	24
line	22
str	19
input	14
map	11
request	11
footprint	9
ticker	9
(remaining namespaces)	smaller

Kinds covered

function · series/var · constant · declaration · control · operator · type system · literal · semantics · export · import

Internals

Path	Role
docs/pine_v6_full_surface_inventory.md	Full tables + architecture diagrams
scripts/regenerate_inventory_summary.py	Regeneration helper
scripts/generate_builtin_metadata.py	LSP metadata from live surface
Evaluator _build_builtin_map()	Source of dispatch truth

How to use the full file

1. Open docs/pine_v6_full_surface_inventory.md in the repo or Sphinx/site mirror.
2. Search by namespace (## ta , strategy , ...).
3. Trust **dispatch** source rows for “is there a callable?”; read **Notes** for mock/feed caveats.
4. After large builtin PRs, regenerate summary counts before quoting them publicly.



Invariants & edge cases

- 1. **Inventory size is a feature** — summarization must not drop  rows when claiming coverage.
- 2. **Metadata presence ≠ semantic completeness.**
- 3. **Partial heuristic count (8)** is docstring-based — real partials may be higher.
- 4. **Language constructs** (control/export) may be parser-complete without every runtime edge.

Worked examples

Quote coverage honestly

“Against the public Pine v6 function reference (434 symbols), pynescrypt registered **0 missing** dispatch entries as of 2026-07-25; 640 total callables across 60 namespaces. See `docs/pine_v6_full_surface_inventory.md`.”

Programmatic check (sketch)

```
# regenerate after evaluator changes
python scripts/regenerate_inventory_summary.py
```

Failure modes

Mistake	Consequence
Pasting 134KB inventory into Mintlify page	Unreadable docs; broken UX
Citing counts without date	Stale marketing
Equating dispatch hit with TV gold	False precision claims

See also

- [Implementation status](#)
- [Missing features](#)
- [Compatibility](#)
- [Runtime builtins](#)

NUMERICAL VALIDATION

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title: "Numerical validation" description: "Methodology and error bounds for PYNE technical and math builtins versus reference Pine Script™ implementations."

NUMERICAL VALIDATION

Abstract

PYNE's technical and mathematical builtins are validated for **numerical agreement** with reference Pine Script™ implementations under IEEE 754 realities. The long-form report (docs/numerical_validation_report.md , v1.0, Nov 2025) documents methodology, per-family error tables, and edge cases.

Headline (from that report): max observed relative error **0.0022%** (ADX, extreme volatility); average error < **0.0005%**; deterministic ops (SMA, OBV, discrete math) often **exact**.



Treat percentages as **report-era measurements**, not eternal SLAs — re-validate after changing smoothers or bar-mode semantics.

Conceptual model

```
sequenceDiagram
    participant D as OHLCV dataset
    participant TV as Reference Pine
    participant PY as PYNE evaluator
    participant C as Comparator
    D->>TV: same bars
    D->>PY: same bars
    TV->>C: series out
    PY->>C: series out
    C->>C: abs / rel error stats
```

Interface surface

Acceptable thresholds (report)

Band	Relative error	Verdict
Excellent	< 0.001%	Pass
Good	< 0.01%	Pass
Acceptable	< 0.1%	Review
Unacceptable	≥ 0.1%	Fail

Metric definitions

- **Absolute error:** $|\text{ref} - \text{pyne}|$
- **Relative error:** $|\text{ref} - \text{pyne}| / |\text{ref}| \times 100\%$ (guard zero refs)
- **Max / mean / std** over bars and scenarios

Category highlights

Family	Notes
Moving averages	SMA exact; EMA/WMA/... tiny FP error
Oscillators	RSI/stoch slightly higher smoothing error still $\ll 0.01\%$
Trend	ADX highest in report (0.0022% max)
Volume	OBV exact cumulative; MFI inherits typical-price FP
Stats	percentrank exact; variance/stddev excellent
Math	Discrete exact; transcendentals $\sim 1\text{e-}15$ class

Dual-host formula alignment (interpret ↔ compile)

Where fixed, these kernels **share one formula contract** across interpret (evaluator / incremental) and compile (`numba_builtins`) so plot series match bit-identical or $\text{maxdiff} \approx 0$ on dual-run goldens:



Builtin	Shared contract (summary)
<code>ta.rsi</code>	Wilder: SMA seed of first <code>period</code> deltas, then RMA gain/loss; first valid at bar <code>period</code>
<code>ta.roc</code>	Standard lookback % change; <code>na</code> until lookback (no early <code>0.0</code>)
<code>ta.wma</code>	Full non- <code>na</code> window required; no partial-window reweight (nested ROC/WMA safe)
<code>ta.cum</code>	Running sum; NaNs as 0; user series (<code>ad = ta.cum(...)</code>) must not alias builtin A/D
<code>ta.highestbars</code> / <code>lowestbars</code>	Negative bars-back offset; short/all-NaN → <code>-1</code> ; oldest extreme on ties

Goldens: `tests/test_compiler_numba.py` (`TestInterpCompilePlotParityFixes`) and `scripts/compare_interp_compile.py` . Residual dual-host drift (if any) tends to cluster in **nested EMA seed families** (DEMA/TEMA SMA-seed vs first-value) rather than these aligned kernels — see [Compatibility](#).

Error distribution (report aggregate)

Error range	Share (approx.)
Exact 0	~38%
< 0.0001%	~46%
0.0001–0.001%	~14%
0.001–0.01%	~2%
≥ 0.1%	0%

Internals

Path	Role
<code>docs/numerical_validation_report.md</code>	Full tables + bias analysis
<code>tests/test_ta_indicators_*.py</code>	Executable TA tests
<code>tests/test_indicators.py</code> / builtins tests	Additional numeric guards
<code>tests/test_compiler_numba.py</code>	Interpret ↔ compile formula goldens (RSI/ROC/WMA/...)
<code>scripts/compare_interp_compile.py</code>	Corpus plot-series parity harness
Bar-mode vs list-mode (<code>_pine_bar_mode</code>)	Scalar-per-bar vs full series — compare like with like

Methodology (report)

1. Generate synthetic 1k-bar OHLCV across regimes (trend, range, vol, gaps)
2. Validate with real histories (equities, crypto, FX samples)
3. Export reference outputs; run PYNE; element-wise compare
4. Search systematic bias; inspect tails

Invariants & edge cases

1. **na propagation** must match Pine truthiness — numerical compare should mask `na` pairs.
2. **Bar-mode scalars** vs full-series list mode can look like “bugs” if misaligned.
3. **Seeded mocks** (`request.seed`) stabilize stochastic feeds for regression.



- 4. **Recursive smoothers** accumulate FP differently across languages; bounds > bits.
- 5. **Extreme prices** (near-zero, huge) tested; still within excellent band in report.

Worked examples

Relative error helper

```
def rel_err(a: float, b: float) -> float:
    if a == 0 and b == 0:
        return 0.0
    denom = abs(a) if a != 0 else abs(b)
    return abs(a - b) / denom * 100.0
```

Run TA tests

```
pytest tests/test_ta_indicators_1.py tests/test_ta_indicators_2.py -q
```

Failure modes

Symptom	Investigation
Sudden ADX drift	Check Wilder smoothing / seed bars
SMA not exact	Off-by-one window or float input casts
Good unit tests, bad TV export compare	Timezone/session alignment of bars
Only bar 0 differs	Warm-up / <code>na</code> initialization

See also

- [Compatibility](#)
- [Runtime technical builtins](#)
- [Implementation status](#)

ROADMAP

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title: "Roadmap" description: "PYNE forward plan — dual-host residual, runtime residual, plot parity harness; H2/T1/F2/L2 landed (2026-08)."

ROADMAP

Abstract

The roadmap is a **living prioritization document**, not a contract. Source: `docs/ROADMAP.md` (updated **2026-08-19**). Core v6 language/builtins are essentially closed. Product **warm-compile (H2)**, **series caps (T1)**, **pending-fill (F2)**, **L2 webhooks**, **corpus Runtime residual (C1)**, **package Runtime SoT (H1 largely done)**, **interpret Round 9 (0.3.10)**, and **compile object-mode residuals (0.3.16–0.3.17)** have landed (set01–04 parse **99.96%**, Runtime interpret



100% excl. intentional demos). What remains: **H1 worker-only extras**. **P1p** official TV builtins are **0 MISMATCH** (optional set0x community corpus unmeasured). Incremental `nvi` / `pvi` (**T2**) and Supertrend goldens (**F1**) have landed.

Conceptual model

```
P0 docs honesty → P1 dual-host H1 ✓ (package SoT + shims + worker thin wrap)
  → P1 plot parity residual (P1p)
  → H2 warm compile ✓ · T1 series caps ✓ · F2 pending-fill ✓ · C1 corpus ✓
    ↘ T2 nvi/pvi ✓ · F1 Supertrend goldens ✓
      → P6 long-horizon (L1 / L3; L2 ✓)
```

```
flowchart LR
    DONE[Core parser + evaluator] --> H1[H1 package SoT ✓]
    DONE --> P1p[P1p plot parity residual]
    C1[C1 corpus set01-04 ✓] --> DONE
    H2[H2 warm compile ✓] --> OPS[Deploy SLOs]
    T1[T1 series caps ✓] --> PERF[Bar-loop hygiene]
    L2[L2 webhooks ✓] --> HOST[Pro + edge export]
```



Interface surface

Status snapshot

Component	Status
Parser ANTLR4	✓ v5/v6 + multiline / export const
Evaluator	✓ incl. full var/varip, ReAssign
Builtins / TA / collections	✓ broad (0 missing vs public TV ref list)
Strategy events + broker (OCA, commission, risk)	✓
Numba + object-mode compile	✓ MVP+ (disk IR cache, mode=auto , time_arr , cache recovery; 0.3.16–0.3.17 UDT/switch/matrix/drawing/round/color residuals)
Pro API + auth + Docker	✓ (mode default auto ; targets api / api-dev / lsp / cli)
PyPI hoox-pyne	✓ live 0.6.7 (pypi.org/project/hoox-pyne); CLIs pyne / pyne-lsp
CLI surface (check / format / compile / run / prewarm /...)	✓ + Nuitka binaries + CLI Docker image
Alert engine + L2 webhooks	✓ dual-host export + outbound webhooks
Drawing max_*_count GC	✓ package + Pro API + AXIS Pyodide
Package Runtime SoT (pynescript.runtime)	✓ 0.3.4 — Pro API + worker thin wrap
Strategy exits (from_entry , qty_percent , trail, risk cascade)	✓ 0.3.4
request.security HTF simple TA + foreign-na	✓ same-symbol OHLCV/HTF allowlist; foreign → na
Free-tier Pro API guards	✓ bar/script caps, rate, concurrency, mock-only free data
Dual-host TA goldens (ATR / Supertrend / Keltner)	✓ first-party + test_ta_incremental CI
Interpret dispatch + unused derived skip + volume inc (Round 9)	✓ 0.3.10 — bench @ 2000 bars: minimal 2.7×, ta_sma 2.0×, ta_combo 1.45×
Series caps (PYNE_SERIES_CAP)	✓ T1
Warm-compile product path	✓ H2 (SLOs, prewarm, deploy IR cache)
Pending-fill averaging (pyramiding ≤ 0)	✓ F2
Interpret ↔ compile plot parity	⚙ harness + goldens landed; residual MISMATCH tail
Runtime timeout_seconds	✓ 0.3.6 — wall-clock circuit breaker on interpret
UDT array.binary_search* sort_field	✓ 0.3.6 — interpret + compile (Pine August 2026)
Runtime.run(..., libraries=) / POST /run libraries	✓ 0.3.7+ — AXIS import ns/Name/ver ; export finalize in 0.3.8; auto fallback keeps libs
pyne-worker 0.6.0	✓ edge /run ; vendored engine is inconsistent (__about__.py 0.3.8, dist-info 0.4.4) and lags PyPI 0.6.7
PyneTS (@hoox-sh/pynets)	✓ standalone TS library 0.2.0 — interpret + JS compile; Python remains oracle
pyne-agent-worker	✓ sister NL authoring host (POST /v1/chat)



Component	Status
Full TV platform identity	⚙️ out of scope

Completed themes (do not re-plan as greenfield)

- Error message / typing hygiene
- `%%pinescript` Jupyter magic + sample data helpers
- `pynscript lint`
- Yahoo / AlphaVantage / CCXT + realtime datafeed wiring
- StrategyEvent emission + parity fixtures
- Colocated TS extra tool extracted (0.3.7). Product TypeScript library is **PyneTS**.
- Alert engine + dual-host export + **L2 webhooks**
- Round 7: series caps, incremental TA (bb/kama/cmo/stochrsi), parse cache, warm compile, pending-fill, drawing GC
- Round 9 / **0.3.10**: interpret dispatch inlining, unused derived OHLCV skip, incremental `obv` / `wad` / `cmf` / `klinger`

Open backlog (IDs stable)

ID	Item	Pri	Status
H1	Dual-host Runtime (package SoT + backend shims + pyne-worker thin wrap  ; residual = worker-only extras)	P1	 largely done
H2	Product warm-compile (SLOs, prewarm API/CLI, IR cache on in deploy)	P1	
C1	Corpus set01–04 residual (parse 99.96%; Runtime 100% excl. EXPECTED)	P1	 2026-08-09
P1p	Compile/interpret plot parity residual	P1	 first-party + official TV builtins 0 MISMATCH ; optional set0x unmeasured
T1	Cap <code>current_series</code> to <code>max_bars_back</code> / <code>_SERIES_MAX</code>	P2	 <code>PYNE_SERIES_CAP</code> default ON
T2	Incremental TA for remaining heavy kernels	P2	 R7 + 0.3.10 volume inc + 0.3.11 nvi/pvi + 0.3.12 aroon/dpo/donchian/kst
F1	ATR Wilder / TV supertrend re-baseline only with dedicated goldens	P2	 <code>mid±factor·ATR</code> locked; TV ratchet out of scope
F2	Pending-fill averaging when pyramiding ≤ 0	P2	
L1	v5 ↔ v6 converter maturity	P3	open
L2	Webhook alerts productization	P3	 Pro + pyne-worker
L3	Legacy TS Worker builtin parity	P3	sister <code>hoox-sh/pine-worker</code> only — not a product-docs target; new TS work is PyneTS
B1	Real (non-mock) <code>request.*</code> market data	—	⚙️ by design; foreign-na policy landed

Longer horizon (not next): ML wrappers, automatic refactor, parallel/distributed eval experiments.



Priority recommendation

Horizon	Focus
Short	H1 worker extras (logs/profile); optional set0x P1p sweep
Medium	Further nested incremental TA where profiled
Long	L1 converter maturity; TypeScript library work is PyneTS (L2 )

Landed residual notes (2026-08)

- **Plot parity:** Always-on smoke in `tests/test_interp_compile_parity.py` . Full corpus compare: `python scripts/compare_interp_compile.py` → `.cache/interp_compile_parity.json` . Flags `--ignore-hline-keys` / `--ignore-fill-keys` drop structural residuals.
- **Corpus residual (C1):** set01–04 parse **99.96%** (2476/2477); Runtime interpret **100%** excl. EXPECTED_FAIL (2466 OK + 11 intentional demos); set01 **249/249**. Third-party corpora are not shipped in git.
- **Foreign-na:** `request.security` on foreign/complex expressions → `na` both backends; `ChartOHLCPProvider` refuses non-chart symbols.

Internals

Path	Role
<code>docs/ROADMAP.md</code>	Canonical roadmap prose
<code>docs/PROGRESS_REPORT.md</code>	Historical completion narrative
<code>docs/COMPILER_PLAN.md</code>	Compile/numba plan
<code>scripts/compare_interp_compile.py</code>	Plot parity harness
<code>.opencode/plans/</code>	Engineering plans (strategy events, consolidation)
sister <code>hoox-sh/pyne-worker</code>	Python edge evaluate host — docs
sister <code>hoox-sh/pyne-agent-worker</code>	NL authoring host — docs
<code>pynets/</code> submodule	TS library pin v0.2.0 (interpret + JS compile); Python remains oracle

Invariants & edge cases

1. **Roadmap dates lag code** — always diff against `missing_features.md` and tests.
2. **Sister repos** (pyne-worker, pyne-agent-worker) may absorb “integration” items.
3. **AXIS is a separate product tree** (`docs/axis`) — chart UX roadmap is not PYNE-core.
4. Avoid reintroducing deleted backups (`builder.py.bak` , etc.) as “plan items.”
5. Do **not** re-open **H1 / H2 / T1 / T2 / F1 / F2 / L2 / C1** as greenfield — they are marked  (H1 residual = worker extras only; P1p is corpus tail) in `docs/ROADMAP.md` (2026-08-16).



Worked examples

Track a roadmap item with a test

```
# pick a gap, write failing test first, implement, update missing_features.md
pytest tests/test_v6_features.py -k footprint -v
```

Plot parity smoke

```
pytest tests/test_interp_compile_parity.py -q
python scripts/compare_interp_compile.py # opt-in full compare
```

TypeScript library gate (PyneTS)

```
# standalone checkout – not present as pine-worker/ in this repo
cd ../pynets && bun test && bun run typecheck
```

See [PyneTS](#).

Failure modes

Anti-pattern	Why it hurts
Planning features already 	Duplicated work (H2/T1/F2/L2 are done)
Roadmap without tests	Unverifiable “done”
Ignoring by-design limits	Infinite platform chase

See also

- [Missing features](#)
- [Implementation status](#)
- [Interpret ↔ compile parity](#)
- [Alerts](#)
- [pyne-worker](#)
- [pyne-agent-worker](#)
- [Contributing](#)

CONTRIBUTING

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title: "Contributing" description: "Hard constraints, workflow, and style rules for contributors and agents working on PYNE."

CONTRIBUTING

Abstract

Contributions to PYNE are welcome when they respect the repo's **mechanical invariants**: generated code boundaries, src-layout packaging, dual console scripts, test corpus costs, and encryption secrets. This page distills `CONTRIBUTING.md` and `AGENTS.md` into an operational contract for humans and agents.



Conceptual model

```
flowchart LR
    FORK[Fork + branch] --> INST[make install / hatch]
    INST --> TEST[tests green]
    TEST --> EDIT[change code]
    EDIT --> LINT[ruff + mypy]
    LINT --> PR[Pull request]
```

Interface surface

Baseline workflow

1. Fork and branch from `main`
2. Install: `make install` or hatch envs
3. Run tests: `make test` / `hatch run test:test`
4. Implement with tests (prefer TDD for non-trivial behavior)
5. Lint/format: `make lint` / `make fmt` or `hatch run lint:style + lint:typing`
6. Open a PR with rationale + test evidence

Official short form also lives in root `CONTRIBUTING.md`.

Everyday commands

Goal	Command
Editable install + LSP	<code>make install</code>
Full tests	<code>make test</code>
LSP only	<code>make test-lsp</code>
Backend only	<code>make test-backend</code>
Lint / format	<code>make lint</code> / <code>make fmt</code>
Fast build sanity	<code>make build-check</code>
API	<code>make run</code>
LSP process	<code>make run-lsp</code>

Hard constraints

Grammar & generated code

- **Edit only** `src/pynescript/ast/grammar/antlr4/resource/*.g4` for grammar work
- **Never hand-edit** `.../antlr4/generated/` or `.../asdl/generated/`
- Regenerate via hatch `lint:gen-parser` / project scripts; see grammar guides under `.opencode/context/`

No stale backups

Do not recreate removed backups such as `builder.py.bak` or `evaluator/builtins/technical_refactored.py`. Live `builder.py` and `technical.py` are sole sources of truth.



Future annotations

Every new Python file must start with:

```
from __future__ import annotations
```

Enforced by ruff isort `required-imports`.

Console scripts

Preferred	Alias	Role
<code>pyne</code>	<code>pynescrypt</code>	Click CLI
<code>pyne-lsp</code>	<code>pynescrypt-lsp</code>	Language Server (pygls)

Import package remains `pynescrypt`. Do not conflate CLI and LSP entrypoints in docs, packaging, or process managers.

Optional example-script fixture

`tests/conftest.py` only expands `pinescript_filepath` when `--example-scripts-dir` points at a local directory of `*.pine` files. **No third-party corpus is shipped** in the repository.

`tests/data/library/` is reference material, not auto-parametrized.

Builtin metadata

`builtin_metadata.json` is **generated from code**. After adding builtins:

```
python scripts/generate_builtin_metadata.py
```

Do not hand-maintain the catalog long-term.

Interpret / compile parity

When changing `pynescrypt.runtime.Runtime`, the compiler, or plot/series export, check **series parity** between `mode="interpret"` and `mode="compile"`:

```
# Local corpus harness (default 50 scripts × 1000 bars)
python scripts/compare_interp_compile.py --bars 1000 --limit 50

# Broader pass from a path list; ignore one-sided hline/fill keys
python scripts/compare_interp_compile.py --file-list path.txt --limit 0 --workers 4 --timeout-sec 3
```

Always-on unit coverage lives in `tests/test_interp_compile_parity.py` (harness helpers + a small smoke subset). Optional longer path:

```
pytest tests/test_interp_compile_parity.py -m interp_compile_full
# or: PYNE_INTERP_COMPILE_FULL=1 pytest tests/test_interp_compile_parity.py
```

Report artifact: `.cache/interp_compile_parity.json`. Value mismatches on shared `series` keys are regressions. First-party `hline` / `fill` / `bgcolor` / `plotshape` / `barcolor` / `plotarrow` / `plotbar` / `plotcandle` keys match (0.3.12); `drawings` is geometry-only. `--ignore-hline-keys` / `--ignore-fill-keys` are optional for leftover



corpus noise. End-user mode semantics: [Evaluate scripts — Runtime modes](#).

Secrets & crypto

- `scripts/build/.metadata.key` is **gitignored**
- CI must supply stable `CRYPTO_KEY` from `METADATA_KEY` / `_METADATA_KEY` for reproducible `.enc` blobs
- Never commit Fernet keys or API admin tokens

Internals (where to change what)

Want to...	Look at
Parse / unparse	<code>src/pynescrypt/ast/helper.py</code>
Add builtin	<code>src/pynescrypt/ast/evaluator/builtins/<ns>.py</code> + metadata generator
LSP feature	<code>src/pynescrypt/langserver/features/</code> + <code>server.py</code>
CLI subcommand	<code>src/pynescrypt/__main__.py</code>
Grammar	<code>resource/*.g4</code> then regenerate
Pro API route	<code>backend/api/preview.py</code> , <code>backend/app.py</code>
Runtime modes / compile fallback	<code>src/pynescrypt/runtime/host.py</code> (<code>Runtime.run</code>); <code>backend/runtime.py</code> is a shim
Interp/compile series parity	<code>scripts/compare_interp_compile.py</code> , <code>tests/test_interp_compile_parity.py</code>
TS library	sister repo hoox-sh/pynets / <code>@hoox-sh/pynets</code>

Style

- **Ruff** line length 120; broad rule set in `pyproject.toml`
- **Black** target `py310` (via toolchain config)
- **Mypy** strict-ish; exemptions for generated grammar, `evaluator.builtins.*` , tests
- Voice for docs: academic nerdy-cool per `docs/WRITING.md` — no empty marketing adjectives

Release (this repo)

Match root `CONTRIBUTING.md` and `.github/workflows/publish.yml` / `release.yml` / `ghcr.yml` :

1. Bump `__version__` in `src/pynescrypt/__about__.py` and update `CHANGELOG.md` .
2. Align `vscode-extension/package.json` when shipping the VSIX together.
3. Ensure CI is green on `main` .
4. Tag and push: `git tag vX.Y.Z && git push origin vX.Y.Z` .
5. **Publish** (`publish.yml`) builds sdist/wheel (`hoox_pyne-*.whl`) and uploads to PyPI (`PYPI_API_TOKEN` or Trusted Publishing OIDC, environment `pypi`).
6. **Build & Release** (`release.yml`) attaches Nuitka CLI/LSP binaries + VSIX to the GitHub Release.
7. **GHCR** (`ghcr.yml`) publishes `ghcr.io/hoox-sh/pyne/{cli,lsp,api}:X.Y.Z` on `v*` tags.

Dry-run PyPI: Actions → **Publish** → Run workflow → `dry_run=true` . Local: `python -m build && twine check dist/*` .



AXIS charting releases live in [hoox-sh/axis](#) (historical fork: [jango-blockchained/axis](#)).

PyneTS contributions

- Work in [hoox-sh/pynets](#), not a `pine-worker/` directory in this repo
- Bun for install / test / typecheck; `bun run generate` after PYNE `.g4` changes
- Python remains the oracle — see [PyneTS parity](#)
- Do not copy PyneTS sources into this tree; work in the standalone [hoox-sh/pynets](#) repo / `@hoox-sh/pynets` package
- Python Runtime remains the oracle.

Failure modes for contributors

Mistake	Fallout
Editing generated ANTLR/ASDL	Overwritten / review hell
Skipping corpus cost awareness	10-minute “unit” tests
Hand-editing metadata JSON	Drift from dispatch
Changing compile/export without parity check	Silent chart/series drift vs interpret
Committing <code>.metadata.key</code>	Secret leak; rotate immediately
PR without lint	CI red on ruff/mypy

See also

- [Local development](#)
- [CI](#)
- [PyneTS](#)
- [pyne-worker](#)
- [pyne-agent-worker](#)
- [Roadmap](#)
- Root `AGENTS.md` , `CONTRIBUTING.md` , `CODE_OF_CONDUCT.md`